

# Talha Jobayer Zihan

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## PROFESSIONAL SUMMARY

Software engineer and CSE graduate from RUET with full-stack experience in React, FastAPI, and PHP/MySQL, plus hands-on knowledge of integrating AI into real applications, including RAG pipelines (LangChain, FAISS) and transformer-based NLP models. Skilled in Python, C++, and SQL, and in deploying with Docker, Vercel, and Render. Passionate about building scalable, production-ready software with intelligent features.

## EXPERIENCE

- Research Intern – ELITE Research Lab** Jun 2026 – Present
- Conducting R&D in **AI and Machine Learning**: literature review, model development, experimentation, and research-oriented problem solving alongside experienced researchers.
- Industrial Attachment, Frontier Semiconductor BD Ltd** Dhaka | 2 Weeks
- Gained hands-on exposure to semiconductor fabrication processes and quality control.
- Industrial Training, Netro Systems** Rajshahi | 2 Weeks
- Worked with industry-standard software development tools; collaborated with developers on real-world projects.

## PROJECTS

- Cross-Domain NIDS – Network Intrusion Detection Platform** [GitHub](#)
- Built a full-stack NIDS with **React, TypeScript, Vite, FastAPI, and PyTorch** for cross-domain PCAP/PCAPNG analysis, integrating **Lopez17CNN** and **BiC (Bias Correction)** for transfer-based detection.
  - Implemented **Target Domain Neighborhood Refinement (TDNR)** and an analyst dashboard (PCAP upload, threat distribution, alerts); deployed on **Vercel** and **Render**.
- CareerVector – RAG-Based CV Analyzer** [GitHub](#)
- Built a RAG pipeline with **LangChain, FAISS, and OpenAI embeddings** to match CVs against job descriptions with sub-second FAISS IVF retrieval.
  - Cut query latency by **80%** via keyword-based section filtering; improved top-1 precision by **35%** over a TF-IDF baseline.
- Bangla Sarcasm Detection using Transformers** [Kaggle](#)
- Built a **BanglaBERT** classifier reaching **88% F1** on a balanced 10k-sample dataset; used **Mixup, FGM adversarial training, and R-Drop** to reduce overfitting by **15%**, with **OneCycleLR** converging in 3 fewer epochs.
- Image Classifier using CNN** TensorFlow / Keras
- Trained a 3-layer CNN with batch normalization on MNIST (60k samples), achieving **95% test accuracy**.
- Car Rental System Web App** [GitHub](#)
- Built a car rental management system with **PHP, MySQL, HTML/CSS, and JavaScript**: user dashboard (browsing, booking history), admin panel, real-time booking with date validation, and secure authentication.

## RESEARCH WORK

- Fast Gradient Method (FGM) Fine-Tuning for Bengali Sarcasm Detection** [IEEE Xplore Jan 2026](#)
- Proposed FGM adversarial fine-tuning on BanglaBERT; published and presented at ICECTE 2026 (Certificate of Contribution).
- Mental Health Text Classification via Data Augmentation and Transformer Model Ensemble** [IEEE Xplore](#)
- Improved mental health text classification using **data augmentation and transformer model ensembling**.

## EDUCATION

- Rajshahi University of Engineering and Technology (RUET)** 2022 – 2026
- B.Sc. in Computer Science and Engineering CGPA: 3.64 / 4.00
- New Govt. Degree College, Rajshahi** 2018 – 2020
- Higher Secondary Certificate (HSC) GPA: 5.00 / 5.00

## TECHNICAL SKILLS

**Programming:** Python, C++, C, Java, SQL, PHP

**AI / ML:** RAG, LangChain, FAISS, Transformers, BanglaBERT, CNNs, TensorFlow, PyTorch, NumPy, Pandas

**Web / Backend:** React, Next.js, TypeScript, Vite, FastAPI, Tailwind CSS, HTML/CSS, JavaScript

**Tools:** Git, GitHub, Kaggle, Linux, MySQL, Arduino, Vercel, Render, Docker

## CERTIFICATIONS

[Machine Learning in Production](#) (DeepLearning.AI) • [NLP in TensorFlow](#) (DeepLearning.AI) • [Deep Learning & Reinforcement Learning](#) (IBM) • [Machine Learning with Python](#) (IBM) • [Intro to Neural Networks and PyTorch](#) (IBM) • PHP (Laravel) Training – RUET CSE, EDGE Project (80 hrs) • [Agents Course](#) (Hugging Face; smolagents, LangGraph, LlamaIndex, RAG, multi-agent systems)

## LEADERSHIP & VOLUNTEERING

- Officer – Marketing, TEDxRUET** 2024 – 2025
- Co-organized a TEDx event with speakers from diverse fields; managed marketing outreach and audience engagement.